

Exact Asymptotic of the Lower Bound

May 30, 2026

1.1 Derivation of the Recurrence

Recall that our shifted yield recurrence is

$$Y_{n+1} \stackrel{d}{=} I^{(n)} \left(Y_{U_n}^{(1)} + Y_{n-1-U_n}^{(2)} + 1 \right) + J^{(n)} \max \left(Y_{U_n}^{(1)}, Y_{n-1-U_n}^{(2)} \right), \quad Y_1 = Y_2 = 0.$$

Taking expectation on both sides, and letting $y_n = \mathbb{E}[Y_n]$, we have

$$y_{n+1} = \frac{2u}{n} \sum_{i=0}^{n-1} y_i + \frac{v}{n} \sum_{i=0}^{n-1} \mathbb{E} \left[\max \left(Y_i^{(1)}, Y_{n-1-i}^{(2)} \right) \right] + u, \quad n \geq 1, \quad y_0 = y_1 = 0.$$

Since $\max(Y_i, Y_{n-1-i}) \geq Y_i$ and $\max(Y_i, Y_{n-1-i}) \geq Y_{n-1-i}$, we have

$$\mathbb{E}[\max(Y_i, Y_{n-1-i})] \geq \max(y_i, y_{n-1-i}).$$

Hence, for every $n \geq 1$, we have

$$y_{n+1} \geq \frac{2u}{n} \sum_{i=0}^{n-1} y_i + \frac{v}{n} \sum_{i=0}^{n-1} \max(y_i, y_{n-1-i}) + u.$$

Define (y'_n) such that $y'_0 = y'_1 = 0$ and for every $n \geq 1$,

$$y'_{n+1} = \frac{2u}{n} \sum_{i=0}^{n-1} y'_i + \frac{v}{n} \sum_{i=0}^{n-1} \max(y'_i, y'_{n-1-i}) + u.$$

Then $y_n \geq y'_n$ for every $n \geq 0$. Finally, we homogenize the recurrence by defining

$$x_n = \frac{2u + v - 1}{u} y'_n + 1$$

for every $n \geq 0$. We have $x_0 = x_1 = 1$ and for every $n \geq 1$. The recurrence of (x_n) is

$$x_{n+1} = \frac{2u}{n} \sum_{i=0}^{n-1} x_i + \frac{v}{n} \sum_{i=0}^{n-1} \max(x_i, x_{n-1-i}), \quad n \geq 1, \quad x_0 = x_1 = 1. \quad (1.1)$$

We consider the case where $2u + v > 1$.

1.2 Analysis of the Recurrence

Throughout this section, let α be the unique solution of the equation

$$\alpha + v2^{-\alpha} = 2(u + v) - 1. \quad (1.2)$$

Note that $\alpha > 0$ since otherwise, we would have $2u + v = 1$.

Proposition 1. *The sequence (x_n) is increasing.*

Proof. We have $x_0 = x_1 = 1$ and $x_2 = 2u + v$. Assume that $x_{i+1} \geq x_i$ for all $0 \leq i \leq n-1$ with $n \geq 2$. We have

$$\begin{aligned} nx_{n+1} &= 2u \sum_{i=0}^{n-1} x_i + v \sum_{i=0}^{n-1} \max(x_i, x_{n-1-i}) \\ &= 2u \sum_{i=0}^{n-1} x_i + 2v \sum_{i=\lfloor (n+1)/2 \rfloor}^{n-1} x_i + v\delta_n x_{\lfloor n/2 \rfloor}, \end{aligned}$$

where $\delta_n = 1$ if n is odd and $\delta_n = 0$ if n is even.

Let us compute $nx_{n+1} - (n-1)x_n$ for $n \geq 2$. Let $k \geq 1$ be an integer.

- If $n = 2k$, we have

$$2kx_{2k+1} - (2k-1)x_{2k} = 2(u+v)x_{2k-1} - vx_{k-1}.$$

- If $n = 2k+1$, we have

$$(2k+1)x_{2k+2} - 2kx_{2k+1} = 2(u+v)x_{2k} - 2vx_k + vx_k = 2(u+v)x_{2k} - vx_k.$$

Collecting two cases, we have

$$\begin{aligned} nx_{n+1} - (n-1)x_n &= 2(u+v)x_{n-1} - vx_{\lfloor (n-1)/2 \rfloor} \\ \iff n(x_{n+1} - x_n) &= 2(u+v)x_{n-1} - x_n - vx_{\lfloor (n-1)/2 \rfloor} \end{aligned}$$

We have

$$x_n = \frac{2u}{n} \sum_{i=0}^{n-2} x_i + \frac{v}{n} \sum_{i=0}^{n-2} \max(x_i, x_{n-2-i}) \leq (2u+v)x_{n-2}.$$

Hence,

$$\begin{aligned} n(x_{n+1} - x_n) &\geq 2(u+v)x_{n-1} - (2u+v)x_{n-2} - vx_{\lfloor (n-1)/2 \rfloor} \\ &\geq 2(u+v)x_{n-1} - (2u+v)x_{n-1} - vx_{n-1} \\ &\geq 0. \end{aligned}$$

□

Proposition 2. *The sequence (x_n) satisfies the recurrence*

$$nx_{n+1} - (n-1)x_n = 2(u+v)x_{n-1} - vx_{\lfloor (n-1)/2 \rfloor}, \quad n \geq 1, \quad (1.3)$$

with $x_0 = x_1 = 1$.

Lemma 1. *For every $n \geq 0$, we have*

$$M_n = \sum_{i=0}^{n-1} \max(i^\alpha, (n-1-i)^\alpha) \leq 2 \int_{n/2}^n x^\alpha dx.$$

Proof. Let $n = 2k$. We have

$$M_{2k} = 2 \sum_{i=k}^{2k-1} i^\alpha \leq 2 \int_k^{2k} x^\alpha dx = 2 \int_{n/2}^n x^\alpha dx.$$

Let $n = 2k + 1$. We have $k^\alpha = 2 \times \frac{1}{2} \times k^\alpha \leq 2 \int_{k+1/2}^{k+1} x^\alpha dx$. Hence,

$$M_{2k+1} \leq 2 \int_{k+1/2}^{k+1} x^\alpha dx + 2 \int_{k+1}^{2k+1} x^\alpha dx = 2 \int_{k+1/2}^{2k+1} x^\alpha dx = 2 \int_{n/2}^n x^\alpha dx.$$

□

Lemma 2. For every $n \geq 1$, we have $x_n \leq Kn^\alpha$ for some $K > 0$.

Proof. We have $x_2 = 2u + v$. Assume that $x_i \leq Ki^\alpha$ for all $1 \leq i \leq n$ with $n \geq 2$ and $K > 0$ is to be determined. For $n \geq 2$, we have

$$\begin{aligned} nx_{n+1} &= 2u + 2u \sum_{i=1}^{n-1} x_i + v \sum_{i=0}^{n-1} \max(x_i, x_{n-1-i}) \\ &\leq 2u + 2uK \int_1^n x^\alpha dx + 2vK \int_{n/2}^n x^\alpha dx \\ &= 2u + \frac{2uK}{\alpha+1} (n^{\alpha+1} - 1) + \frac{2vK}{\alpha+1} (n^{\alpha+1} - (n/2)^{\alpha+1}) \\ &= 2u + \frac{Kn^{\alpha+1}}{\alpha+1} (2u + 2v(1 - 2^{-\alpha-1})) - \frac{2uK}{\alpha+1} \\ &= 2u + Kn^{\alpha+1} - \frac{2uK}{\alpha+1}. \end{aligned}$$

Note that in the sum of maxima, we do not have to isolate $\max(x_0, x_{n-1})$, since it is automatically evaluated as x_{n-1} , which is already included in the integral. Choose $K = \alpha + 1$, we check that the base cases hold

$$\begin{aligned} \alpha + 1 &\geq 1, \\ (\alpha + 1)2^\alpha &= (2(u + v) - v2^{-\alpha}) 2^\alpha = 2^{\alpha+1}(u + v) - v \geq 2u + v. \end{aligned}$$

Therefore,

$$x_{n+1} \leq Kn^\alpha \leq K(n+1)^\alpha.$$

□

Theorem 1. We have $x_n \sim Cn^\alpha$ for some $C > 0$ to be determined and α is the unique solution of the equation

$$\alpha + v2^{-\alpha} = 2(u + v) - 1$$

Proof. Let $z_n = \frac{x_n}{(n-1)^\alpha}$ for $n \geq 2$. We will prove that (z_n) converges by proving that $\sum_{n \geq 0} |z_{n+1} - z_n|$ converges. We have, for $n \geq 5$,

$$\begin{aligned} n^{\alpha+1} z_{n+1} - (n-1)^{\alpha+1} z_n &= 2(u+v)(n-2)^\alpha z_{n-1} - v \left(\left\lfloor \frac{n-1}{2} \right\rfloor - 1 \right)^\alpha z_{\lfloor (n-1)/2 \rfloor} \\ \iff z_{n+1} &= \left(1 - \frac{1}{n} \right)^{\alpha+1} z_n + \frac{2(u+v)}{n} \left(1 - \frac{2}{n} \right)^\alpha z_{n-1} - \frac{v}{n} \left(\frac{\lfloor (n-1)/2 \rfloor - 1}{n} \right)^\alpha z_{\lfloor (n-1)/2 \rfloor}. \end{aligned}$$

We have

$$\begin{aligned} \left(1 - \frac{1}{n}\right)^{\alpha+1} &= 1 - \frac{\alpha+1}{n} + O\left(\frac{1}{n^2}\right), \\ \frac{1}{n} \left(1 - \frac{2}{n}\right)^{\alpha} &= \frac{1}{n} + O\left(\frac{1}{n^2}\right), \\ -\frac{1}{n} \left(\frac{\lfloor (n-1)/2 \rfloor - 1}{n}\right)^{\alpha} &= -\frac{1}{n} \left(2^{-\alpha} + O\left(\frac{1}{n}\right)\right) = -\frac{2^{-\alpha}}{n} + O\left(\frac{1}{n^2}\right). \end{aligned}$$

By the previous lemma, (z_n) is bounded. Therefore,

$$\begin{aligned} z_{n+1} - z_n &= -\frac{\alpha+1}{n} z_n + \frac{2(u+v)}{n} z_{n-1} - \frac{v2^{-\alpha}}{n} z_{\lfloor (n-1)/2 \rfloor} + O\left(\frac{1}{n^2}\right) \\ &= -\frac{2(u+v)}{n} (z_n - z_{n-1}) + \frac{v2^{-\alpha}}{n} (z_n - z_{\lfloor (n-1)/2 \rfloor}) + O\left(\frac{1}{n^2}\right). \end{aligned}$$

Let $\nabla z_n = z_n - z_{n-1}$. Then, there exists a constant $L > 0$ such that for every $n \geq 5$,

$$|\nabla z_{n+1}| \leq \frac{2(u+v)}{n} |\nabla z_n| + \frac{v2^{-\alpha}}{n} \sum_{i=\lfloor (n+1)/2 \rfloor}^n |\nabla z_i| + \frac{L}{(n+1)^2}.$$

Equivalently, for every $n \geq 6$,

$$|\nabla z_n| \leq \frac{2(u+v)}{n-1} |\nabla z_{n-1}| + \frac{v2^{-\alpha}}{n-1} \sum_{i=\lfloor n/2 \rfloor}^{n-1} |\nabla z_i| + \frac{L}{n^2}.$$

Let $u_m = \max_{2^m \leq n < 2^{m+1}} |\nabla z_n|$. Then, for every $n \in [2^m, 2^{m+1})$, we have

$$\begin{aligned} \frac{1}{n-1} |\nabla z_{n-1}| &\leq \frac{1}{n-1} \max_{2^{m-1} \leq i < 2^m} |\nabla z_i| = \frac{1}{n-1} \max(u_{m-1}, u_m), \\ \frac{1}{n-1} \sum_{i=\lfloor n/2 \rfloor}^{n-1} |\nabla z_i| &\leq \frac{n+1}{2(n-1)} \max_{2^{m-1} \leq i < 2^m} |\nabla z_i| \leq \left(\frac{1}{2} + \frac{1}{n-1}\right) \max(u_{m-1}, u_m). \end{aligned}$$

Therefore,

$$\begin{aligned} |\nabla z_n| &\leq \frac{2(u+v)}{n-1} \max(u_{m-1}, u_m) + v2^{-\alpha} \left(\frac{1}{2} + \frac{1}{n-1}\right) \max(u_{m-1}, u_m) + \frac{L}{2^{2m}} \\ &= \left(\frac{\alpha+1}{n-1} + v2^{-\alpha-1}\right) \max(u_{m-1}, u_m) + \frac{L}{2^{2m}}. \end{aligned}$$

Since $v2^{-\alpha} = 2(u+v) - \alpha - 1 < 1$, we can choose m_0 large enough such that for every $m \geq m_0$ and $n \in [2^m, 2^{m+1})$, there exists γ_1 such that $\frac{\alpha+1}{n-1} + v2^{-\alpha-1} < \gamma_1 < \frac{1}{2}$. Taking the maximum over $n \in [2^m, 2^{m+1})$, we have

$$u_m \leq \gamma_1 \max(u_{m-1}, u_m) + \frac{L}{2^{2m}}.$$

If $u_{m-1} > u_m$, then $u_m \leq \gamma_1 u_{m-1} + \frac{L}{2^{2m}}$.

If $u_{m-1} \leq u_m$, then $u_m \leq \frac{L}{2^{2m}(1-\gamma_1)}$.

In both cases, we have

$$\begin{aligned}
u_m &\leq \gamma_1 u_{m-1} + \frac{L}{2^{2m}(1-\gamma_1)} \\
&\leq \sum_{i=m_0+1}^m \gamma_1^{m-i} \frac{L}{2^{2i}(1-\gamma_1)} + \gamma_1^{m-m_0} u_{m_0} \\
&< \frac{L}{(1-\gamma_1)} \sum_{i=m_0+1}^m \gamma_1^{m-i} \left(\frac{1}{4}\right)^i + \gamma_1^{m-m_0} u_{m_0}.
\end{aligned}$$

Choose γ_2 such that $\max\left(\gamma_1, \frac{1}{4}\right) < \gamma_2 < \frac{1}{2}$. We have

$$u_m < \frac{L}{(1-\gamma_1)} \sum_{i=m_0+1}^m \gamma_2^m + \gamma_2^{m-m_0} u_{m_0} \leq L' m \gamma_2^m,$$

for some constant $L' > 0$. Choose γ_3 such that $\gamma_2 < \gamma_3 < \frac{1}{2}$. Then there exists $m_1 > m_0$ such that for every $m \geq m_1$, we have $L' m \gamma_2^m < L' \gamma_3^m$, implying that,

$$u_m < L' \gamma_3^m.$$

Therefore,

$$\sum_{n=0}^{\infty} |\nabla z_n| = \sum_{m=0}^{\infty} \sum_{n=2^m}^{2^{m+1}-1} |\nabla z_n| = \sum_{m=0}^{\infty} 2^m u_m < \sum_{m=0}^{m_1-1} 2^m u_m + L' \sum_{m=m_1}^{\infty} (2\gamma_3)^m < \infty.$$

Hence, (z_n) converges to some constant $C > 0$. Thus, $x_n \sim Cn^\alpha$. □

Existence of a True Power

June 10, 2026

2.1 The Recurrence and Fixed-Point Equation

Let us recall the recurrence for Y_n

$$Y_{n+2} \stackrel{d}{=} I^{(n)} \left(Y_{U_n}^{(1)} + Y_{n+1-U_n}^{(2)} \right) + J^{(n)} \max \left(Y_{U_n}^{(1)}, Y_{n+1-U_n}^{(2)} \right), n \geq 1, \quad Y_1 = 1, Y_2 = 0.$$

Because our analysis concerns only the sequence of marginal distributions $\mathcal{L}(Y_n)$, we can use the same random variables I, J for each n . This reuse does not impact the distributional recurrence, provided that at each step n , the inputs $Y_{U_n}^{(1)}$ and $Y_{n+1-U_n}^{(2)}$ are drawn as conditionally independent copies from the current distribution $\mathcal{L}(Y_n)$, independent of I, J , and U_n . So the recurrence can be rewritten as

$$Y_{n+2} \stackrel{d}{=} I \left(Y_{U_n}^{(1)} + Y_{n+1-U_n}^{(2)} \right) + J \max \left(Y_{U_n}^{(1)}, Y_{n+1-U_n}^{(2)} \right), n \geq 1, \quad Y_1 = 1, Y_2 = 0. \quad (2.1)$$

For the same reason, we set $U_n = \lceil nU \rceil$ for every n , where U is uniformly distributed on $[0, 1]$. Let $X_n = Y_n/n^\alpha$, for some $\alpha \in (0, 1]$. We have the recurrence for X_n as follows

$$X_{n+2} \stackrel{d}{=} I \left(A_n^\alpha X_{U_n}^{(1)} + B_n^\alpha X_{n+1-U_n}^{(2)} \right) + J \max \left(A_n^\alpha X_{U_n}^{(1)}, B_n^\alpha X_{n+1-U_n}^{(2)} \right), n \geq 1, \quad X_1 = 1, X_2 = 0, \quad (2.2)$$

where

$$A_n = \frac{U_n}{n+2} \text{ and } B_n = \frac{n+1-U_n}{n+2}.$$

Suppose that (X_n) converges, then limiting distribution Z of Y_n/n^α satisfies the following fixed-point equation

$$Z \stackrel{d}{=} I \left(U^\alpha Z^{(1)} + (1-U)^\alpha Z^{(2)} \right) + J \max \left(U^\alpha Z^{(1)}, (1-U)^\alpha Z^{(2)} \right), \quad Z^{(1)}, Z^{(2)} \stackrel{\text{iid}}{\sim} Z. \quad (2.3)$$

We aim to find solutions (Z, α) to 2.3 such that $\mathcal{L}(Z) \neq \delta_0$. Let us work with the space $\mathcal{D}_2(1)$ of distributions over $\mathbb{R}_+$ with mean 1 and finite second moment. Recall the p -Wasserstein metric d_p on $\mathcal{D}_2(1)$ for $p \in [1, 2]$

$$d_p(\mu, \nu) = \inf \left\{ \mathbb{E} [(X - Y)^p]^{1/p} : X \sim \mu, Y \sim \nu \right\}.$$

That means for every $X \sim \mu$ and $Y \sim \nu$, we have $d_p(\mu, \nu) \leq \mathbb{E} [(X - Y)^p]^{1/p}$.

We give some properties of the fixed-point equation 2.3. Define the operator $T_\alpha : \mathcal{D}_2(1) \rightarrow \mathcal{D}_2(1)$ according to the right-hand side of 2.3 as

$$T_\alpha(\mu) = \mathcal{L} \left(I \left(U^\alpha Z^{(1)} + (1-U)^\alpha Z^{(2)} \right) + J \max \left(U^\alpha Z^{(1)}, (1-U)^\alpha Z^{(2)} \right) \right), \quad Z^{(1)}, Z^{(2)} \stackrel{\text{iid}}{\sim} \mu. \quad (2.4)$$

Lemma 3. For every $\mu, \nu \in \mathcal{D}_2(1)$ and $\alpha \in (0, 1]$, we have

$$d_2(T_\alpha(\mu), T_\alpha(\nu)) \leq \sqrt{\frac{2(u+v)}{2\alpha+1}} d_2(\mu, \nu) \text{ and } d_1(T_\alpha(\mu), T_\alpha(\nu)) \leq \frac{2(u+v)}{\alpha+1} d_1(\mu, \nu).$$

Proof. Since I and J are Bernoulli, we have $I^2 = I$ and $J^2 = J$. Also, from $I + J \leq 1$, we have $IJ = 0$. Therefore, for every $X, X^{(1)}, X^{(2)} \stackrel{\text{iid}}{\sim} \mu$ and $Y, Y^{(1)}, Y^{(2)} \stackrel{\text{iid}}{\sim} \nu$, we have

$$\begin{aligned} d_2^2(T_\alpha(\mu), T_\alpha(\nu)) &\leq \mathbb{E} \left[I \left(\left(U^\alpha (X^{(1)} - Y^{(1)}) + (1-U)^\alpha (X^{(2)} - Y^{(2)}) \right)^2 \right) \right] \\ &\quad + \mathbb{E} \left[J \left(\max \left(U^\alpha X^{(1)}, (1-U)^\alpha X^{(2)} \right) - \max \left(U^\alpha Y^{(1)}, (1-U)^\alpha Y^{(2)} \right) \right)^2 \right] \\ &:= A + B. \end{aligned}$$

For the first expectation A , we use $\mathbb{E}[X] = \mathbb{E}[Y] = 1$ and get

$$\begin{aligned} A &= u \mathbb{E} \left[U^{2\alpha} (X^{(1)} - Y^{(1)})^2 + (1-U)^{2\alpha} (X^{(2)} - Y^{(2)})^2 \right] \\ &= \frac{2u}{2\alpha+1} \mathbb{E}[(X - Y)^2]. \end{aligned}$$

For the second one, B , we use the inequality

$$(\max(a, b) - \max(c, d))^2 \leq \max(a - c, b - d)^2 \leq (a - c)^2 + (b - d)^2$$

for real numbers a, b, c, d and get

$$\begin{aligned} B &\leq v \mathbb{E} \left[U^{2\alpha} (X^{(1)} - Y^{(1)})^2 + (1-U)^{2\alpha} (X^{(2)} - Y^{(2)})^2 \right] \\ &= \frac{2v}{2\alpha+1} \mathbb{E}[(X - Y)^2]. \end{aligned}$$

Therefore,

$$d_2^2(T_\alpha(\mu), T_\alpha(\nu)) \leq \frac{2(u+v)}{2\alpha+1} \mathbb{E}[(X - Y)^2].$$

Taking the expectation over $X \sim \mu$ and $Y \sim \nu$, we get the desired result. The case for d_1 is similar. $\square$

For $\mu \in \mathcal{D}_2(1)$ and $\alpha \in (0, 1]$, define

$$f(\mu, \alpha) = \mathbb{E}_{Y \sim T_\alpha(\mu)}[Y] = \frac{2u}{\alpha+1} + v \mathbb{E}_{Z^{(1)}, Z^{(2)} \stackrel{\text{iid}}{\sim} \mu} \left[\max \left(U^\alpha Z^{(1)}, (1-U)^\alpha Z^{(2)} \right) \right] \quad (2.5)$$

Lemma 4. For every $\mu \in \mathcal{D}_2(1)$, there exists a unique $\alpha \in (0, 1]$ such that $f(\mu, \alpha) = 1$.

Proof. It is clear that f is strictly decreasing in α . Next, we show that f is continuous. The first term is continuous. We show that the second term is Lipschitz continuous. Let $\alpha, \alpha' \in (0, 1]$

and assume without loss of generality that $\alpha > \alpha'$. We have

$$\begin{aligned}
& \mathbb{E} \left[\max \left(U^\alpha X^{(1)}, (1-U)^\alpha X^{(2)} \right) - \max \left(U^{\alpha'} X^{(1)}, (1-U)^{\alpha'} X^{(2)} \right) \right] \\
& \leq \mathbb{E} \left[\max \left(\left| U^\alpha - U^{\alpha'} \right| X^{(1)}, \left| (1-U)^\alpha - (1-U)^{\alpha'} \right| X^{(2)} \right) \right] \\
& \leq \mathbb{E} \left[\left| U^\alpha - U^{\alpha'} \right| \right] + \mathbb{E} \left[\left| (1-U)^\alpha - (1-U)^{\alpha'} \right| \right] \\
& = 2\mathbb{E} \left[\left| U^\alpha - U^{\alpha'} \right| \right] \quad (U \stackrel{d}{=} 1-U) \\
& = 2\mathbb{E} \left[U^{\alpha'} - U^\alpha \right] \quad (U^\alpha \geq U^{\alpha'}, a.s.) \\
& = \frac{2}{\alpha' + 1} - \frac{2}{\alpha + 1} = \frac{2(\alpha - \alpha')}{(\alpha + 1)(\alpha' + 1)} \\
& \leq 2(\alpha - \alpha').
\end{aligned}$$

Finally, we evaluate f at the endpoints.

$$\begin{aligned}
\lim_{\alpha \rightarrow 0+} f(\mu, \alpha) &= 2u + v \mathbb{E} \left[\max \left(X^{(1)}, X^{(2)} \right) \right] > 2u + v > 1, \\
f(1) &= u + v \mathbb{E} \left[\max \left(U^\alpha X^{(1)}, (1-U)^\alpha X^{(2)} \right) \right] \leq u + v \leq 1.
\end{aligned}$$

□

Consider $\alpha : \mathcal{D}_2(1) \rightarrow (0, 1]$ as a function that maps a distribution μ to the unique α such that $f(\mu, \alpha) = 1$. By Lemma 4, this function is well-defined. That motivates an algorithm to find α^* and the limiting distribution μ^* of X_n as follows.

Algorithm 1

- 1: Initialize $Z_1 = 1$ and α_1 solves $\alpha + v2^{-\alpha} = 2(u + v) - 1$.
 - 2: **for** $k = 1, 2, \dots$ **do**
 - 3: Compute $Z_{n+1} = T_{\alpha_n}(\mathcal{L}(Z_n))$.
 - 4: Find α_{n+1} such that $\mathbb{E}[Z_{n+1}] = 1$ i.e. $f(\mathcal{L}(Z_{n+1}), \alpha_{n+1}) = 1$.
 - 5: **endfor**
-

Using $\max(a, b)^2 \leq a^2 + b^2$, we bound the second moments as follows

$$\begin{aligned}
Z_{n+1}^2 &\leq I \left(U^{2\alpha_n} \left(Z_n^{(1)} \right)^2 + (1-U)^{2\alpha_n} \left(Z_n^{(2)} \right)^2 + 2(U(1-U))^{\alpha_n} Z_n^{(1)} Z_n^{(2)} \right) \\
&\quad + J \left(U^{2\alpha_n} \left(Z_n^{(1)} \right)^2 + (1-U)^{2\alpha_n} \left(Z_n^{(2)} \right)^2 \right) \\
&= (I + J) \left(U^{2\alpha_n} \left(Z_n^{(1)} \right)^2 + (1-U)^{2\alpha_n} \left(Z_n^{(2)} \right)^2 \right) + 2I(U(1-U))^{\alpha_n} Z_n^{(1)} Z_n^{(2)}
\end{aligned}$$

Taking the expectation, we get

$$\mathbb{E} [Z_{n+1}^2] \leq \frac{2(u+v)}{2\alpha_n + 1} \mathbb{E} [Z_n^2] + 2uB(\alpha_n + 1, \alpha_n + 1),$$

where B is the beta function. The function $\alpha \mapsto B(\alpha + 1, \alpha + 1)$ is bounded on $(0, 1]$. In particular, we have $B(\alpha + 1, \alpha + 1) \in [1/6, 1)$. For $\sup_n \mathbb{E}[Z_n^2] < \infty$, at least we need some k such that

$$\frac{2(u+v)}{2\alpha_k + 1} < 1 \iff \alpha_k > u + v - \frac{1}{2}$$

We may choose to impose $\alpha_1 > u + v - \frac{1}{2}$ and use $\alpha_1 > 2u + v - 1$ to derive the condition $u \geq \frac{1}{2}$.

2.2 Existence of a Fixed Point

Let us define the operator $\Phi : \mathcal{D}_2(1) \rightarrow \mathcal{D}_2(1)$ as

$$\Phi(\mu) = T_{\alpha(\mu)}(\mu). \quad (2.6)$$

Let $\mathcal{D}_2(1, M)$ be the subset of $\mathcal{D}_2(1)$ with second moment bounded by M

$$\mathcal{D}_2(1, M) = \left\{ \mu \in \mathcal{D}_2(1) : \int_0^\infty x^2 d\mu \leq M \right\}.$$

Lemma 5. *There exists a real number M such that the operator Φ defined in 2.6 maps $\mathcal{D}_2(1, M)$ into itself.*

Proof. Let $\mu \in \mathcal{D}_2(1)$ have finite second moment. Let $Z \sim \mu$. Follow the steps in the previous section, we have

$$\begin{aligned} \mathbb{E} [\Phi(Z)^2] &\leq \frac{2(u+v)}{2\alpha+1} \mathbb{E} [Z^2] + 2uB(\alpha+1, \alpha+1) \\ &< \frac{2(u+v)}{2(2u+v)-1} \mathbb{E} [Z^2] + 2uB(2u+v, 2u+v) \end{aligned}$$

Choosing

$$M = B(2u+v, 2u+v) \frac{2u(2u-1)}{2(2u+v)-1} \text{ and } \mathbb{E}[Z^2] \leq M,$$

we have $\mathbb{E} [\Phi(Z)^2] \leq M$. Hence, Φ maps $\mathcal{D}_2(1, M)$ into itself. $\square$

We attempt to use the Tychonoff fixed-point theorem (See the Discussion section for why Banach or Schauder fixed-point theorems do not apply here).

Theorem 2 (Tychonoff [4, Cor. 9.6]). *Let $\mathcal{C}$ be a nonempty, compact, convex subset of a locally convex space $\mathcal{M}$. Let $f : \mathcal{C} \rightarrow \mathcal{C}$ be a continuous mapping. Then f has a fixed point.*

The host space $\mathcal{M}$ is the space of signed Borel measures on $\mathbb{R}$, equipped with the topology of weak convergence. Details of $\mathcal{M}$ being locally convex are provided in the Appendix.

Then $\Phi : \mathcal{D}_2(1, M) \rightarrow \mathcal{D}_2(1, M)$.

Lemma 6. *The subset $\mathcal{D}_2(1, M)$ of $\mathcal{M}$ is nonempty, compact, and convex.*

Proof. The set $\mathcal{D}_2(1, M)$ is nonempty since it contains δ_1 . For convexity, let $\mu, \nu \in \mathcal{D}_2(1, M)$. Then for every $\lambda \in [0, 1]$, we have

$$\begin{aligned} \int_0^\infty x d(\lambda\mu + (1-\lambda)\nu) &= \lambda \int_0^\infty x d\mu + (1-\lambda) \int_0^\infty x d\nu = 1, \\ \int_0^\infty x^2 d(\lambda\mu + (1-\lambda)\nu) &= \lambda \int_0^\infty x^2 d\mu + (1-\lambda) \int_0^\infty x^2 d\nu \leq M. \end{aligned}$$

For compactness, we need to show that $\mathcal{D}_2(1, M)$ is tight (cf. Definition ??) and closed. For tightness, we use Markov's inequality. For any $\mu \in \mathcal{D}_2(1, M)$ and any $R > 0$

$$\mu(\{x \geq R\}) \leq \frac{1}{R^2} \int_0^\infty x^2 d\mu \leq \frac{M}{R^2}. \quad (2.7)$$

For closedness, let (μ_n) be a sequence in $\mathcal{D}_2(1, M)$ that converges weakly to μ . We show that $\mu \in \mathcal{D}_2(1, M)$. The boundedness of the second moment follows from a generalized Fatou's lemma [2, Theorem 1.1]

$$\int_0^\infty x^2 d\mu \leq \liminf_{n \rightarrow \infty} \int_0^\infty x^2 d\mu_n \leq M.$$

The function $x \mapsto x$ is uniformly integrable with respect to $\mathcal{D}_2(1, M)$ as follows. For any $\mu \in \mathcal{D}_2(1, M)$ and any $R > 0$

$$\int_R^\infty x d\mu \leq \int_R^\infty x \left(\frac{x}{R}\right) d\mu = \frac{1}{R} \int_R^\infty x^2 d\mu \leq \frac{1}{R} \int_0^\infty x^2 d\mu \leq \frac{M}{R}.$$

which tends to 0 as $R \rightarrow \infty$. Therefore, the first moment converges [1, Theorem 3.5]

$$\int_0^\infty x d\mu_n \rightarrow \int_0^\infty x d\mu.$$

□

Remark. Since $x \mapsto x$ is uniformly integrable with respect to $\mathcal{D}_2(1, M)$, weak convergence in $\mathcal{D}_2(1, M)$ is equivalent to convergence in d_1 [3, Theorem 6.9].

Lemma 7. *The operator Φ defined in 2.6 is continuous on $\mathcal{D}_2(1, M)$ with respect to the weak topology.*

Proof. Let (μ_n) be a sequence in $\mathcal{D}_2(1, M)$ that converges weakly to μ , which is equivalent to that $d_1(\mu_n, \mu) \rightarrow 0$. To show that $\Phi(\mu_n) \xrightarrow{w} \Phi(\mu)$, we will show that $d_1(\Phi(\mu_n), \Phi(\mu)) \rightarrow 0$ by showing that Φ is Lipschitz with respect to d_1 . Let $\mu, \nu \in \mathcal{D}_2(1, M)$. Shorten $\alpha_1 := \alpha(\mu)$ and $\alpha_2 := \alpha(\nu)$. Without loss of generality, assume that $\alpha_1 > \alpha_2$. We have

$$\begin{aligned} d_1(\Phi(\mu), \Phi(\nu)) &= d_1(T_{\alpha_1}(\mu), T_{\alpha_2}(\nu)) \\ &\leq d_1(T_{\alpha_2}(\mu), T_{\alpha_2}(\nu)) + d_1(T_{\alpha_1}(\mu), T_{\alpha_2}(\mu)). \end{aligned}$$

By Lemma 3, we have

$$d_1(T_{\alpha_2}(\mu), T_{\alpha_2}(\nu)) \leq \frac{2(u+v)}{\alpha_2+1} d_1(\mu, \nu) \leq 2(u+v) d_1(\mu, \nu).$$

For the second distance, we let $Y^{(1)}, Y^{(2)} \sim \nu$. Then, following the same steps as in Lemma 4, we have

$$d_1(T_{\alpha_1}(\mu), T_{\alpha_2}(\mu)) \leq 2(u+v)(\alpha_1 - \alpha_2).$$

We have $f(\mu, \alpha_1) = f(\nu, \alpha_2) = 1$. Hence

$$|f(\mu, \alpha_1) - f(\mu, \alpha_2)| = |f(\mu, \alpha_2) - f(\nu, \alpha_2)|.$$

Consider the left-hand side term where we fix the distribution μ . We rewrite f as

$$f(\mu, \alpha) = \frac{2u}{\alpha+1} + v\mathbb{E} \left[U^\alpha Z^{(1)} \mathbf{1}_A + (1-U)^\alpha Z^{(2)} \mathbf{1}_{A^c} \right], \quad Z^{(1)}, Z^{(2)} \stackrel{\text{iid}}{\sim} \mu,$$

where A is the event that $U^\alpha Z^{(1)} \geq (1 - U)^\alpha Z^{(2)}$. Since $\mathbb{P}(U^\alpha Z^{(1)} = (1 - U)^\alpha Z^{(2)}) = 0$, we have

$$\frac{\partial}{\partial \alpha} f(\mu, \alpha) = -\frac{2u}{(\alpha + 1)^2} + v\mathbb{E} \left[\ln(U)U^\alpha Z^{(1)} \mathbf{1}_A + \ln(1 - U)(1 - U)^\alpha Z^{(2)} \mathbf{1}_{A^c} \right].$$

Since $\ln(U) < 0$ and $\ln(1 - U) < 0$, we have for every $\alpha \in (0, 1]$,

$$\left| \frac{\partial}{\partial \alpha} f(\mu, \alpha) \right| \geq \frac{2u}{(\alpha + 1)^2}.$$

By Mean Value Theorem, we have

$$|f(\mu, \alpha_1) - f(\mu, \alpha_2)| \geq \frac{2u}{(\alpha_1 + 1)^2} |\alpha_1 - \alpha_2| \geq \frac{u}{2} |\alpha_1 - \alpha_2|$$

We also have

$$|f(\mu, \alpha_2) - f(\nu, \alpha_2)| \leq \frac{2v}{\alpha_2 + 1} d_1(\mu, \nu) \leq 2vd_1(\mu, \nu).$$

Thus,

$$d_1(\Phi(\mu), \Phi(\nu)) \leq 2(u + v) \left(1 + \frac{4v}{u} \right) d_1(\mu, \nu).$$

□

Theorem 3. *The operator Φ defined in 2.6 has a fixed point in $\mathcal{D}_2(1, M)$. That is, there exists a pair $\alpha^* \in (0, 1]$ and $\mu^* \in \mathcal{D}_2(1, M)$ satisfying (2.3) with $Z \sim \mu^*$.*

Proof. The theorem follows from Tychonoff's fixed-point theorem, Lemma 6, and Lemma 7. □

2.3 Uniqueness of the Fixed Point

For uniqueness, the intuition is as follows. Algorithm 1 only converges if we adjust α_n at each step to ensure that the mean of $\int x d\mu_n = 1$. We started from the most concentrated distribution $\mu_1 = \delta_1$ and increased α_n to fight against the variance of δ_n . Hence, if we fix α^* , then the sequence (μ_n) will converge to δ_0 . Otherwise, if we fix $\alpha = \alpha^* - \epsilon$, then the sequence will diverge to ∞ .

If a single α^* gives two distributions μ_1^* and μ_2^* that are fixed points of T_{α^*} , then from Lemma 3, we have

$$d_2(\mu_1^*, \mu_2^*) \leq \sqrt{\frac{2(u + v)}{2\alpha + 1}} d_2(\mu_1^*, \mu_2^*).$$

Hence $\mu_1^* = \mu_2^*$.

Lemma 8. *Let (α^*, μ^*) be a solution to (2.3). Then $T_{\alpha^*}^n(\delta_1) \xrightarrow{d} \delta_0$.*

2.4 Convergence of the Algorithm

References

- [1] Patrick Billingsley. *Convergence of Probability Measures*. Vol. 100. Wiley Online Library, 1999, pp. 80–120.
- [2] Eugene A Feinberg, Pavlo O Kasyanov, and Nina V Zadoianchuk. “Fatou’s lemma for weakly converging probabilities”. In: *Theory of Probability & Its Applications* 58.4 (2014), pp. 683–689.
- [3] Cédric Villani et al. *Optimal transport: old and new*. Vol. 338. Springer, 2009.
- [4] Eberhard Zeidler. *Nonlinear functional analysis and its applications: I: Fixed-point theorems*. Springer-Verlag New York Inc., 1986.